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JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 26th February Morning Shift

JEE Main Question Paper with Answer Key

26 Feb 2021	90	19	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	21 questions
Physics	28 questions
Out of syllabus	19 questions
Answer key	at the end

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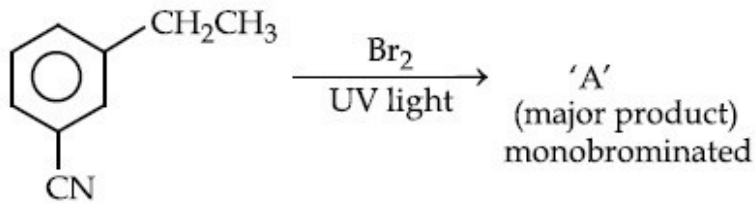
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Chemistry 22 questions

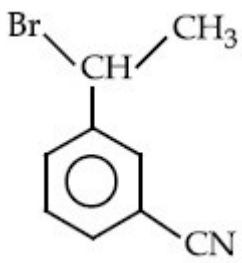
Q1. MCQ (Single Correct Answer)

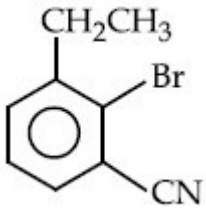
Open solution

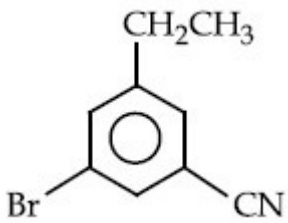
For the given reaction

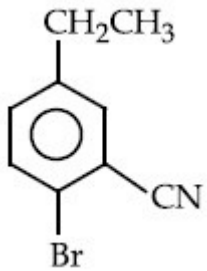


What is 'A'?

(A) 

(B) 

(C) 

(D) 

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Q2. MCQ (Single Correct Answer)

Which of the following is 'a' FALSE statement ?

(A) Carius method is used for the estimation of nitrogen in an organic compound.

(B) Kjeldahl's method is used for the estimation of nitrogen in an organic compound.

(C) Carius tube is used in the estimation of sulphur in an organic compound.

(D) Phosphoric acid produced on oxidation of phosphorus present in an organic compound is precipitated as $Mg_2P_2O_7$ by adding magnesia mixture.

Open solution

Q3. MCQ (Single Correct Answer)

Find A, B and C in the following reactions :

$$NH_3 + A + CO_2 \rightarrow (NH_4)_2CO_3$$

$$(NH_4)_2CO_3 + H_2O + B \rightarrow NH_4HCO_3$$

$$NH_4HCO_3 + NaCl \rightarrow NH_4Cl + C$$

(A) $A - O_2; B - CO_2; C - Na_2CO_3$

(B) $A - H_2O; B - O_2; C - NaHCO_3$

(C) $A - H_2O; B - O_2; C - Na_2CO_3$

(D) $A - H_2O; B - CO_2; C - NaHCO_3$

Open solution

Q4. MCQ (Single Correct Answer)

Match List - I with List - II.

	List I **Electronic configuration**		**List II** $\Delta_i H$ **in kJ** mol^{-1}
(a)	$1s^2 2s^2$	(i)	801
(b)	$1s^2 2s^2 2p^4$	(ii)	899
(c)	$1s^2 2s^2 2p^3$	(iii)	1314
(d)	$1s^2 2s^2 2p^1$	(iv)	1402

Choose the most appropriate answer from the options given below :

(A) (a) → (i), (b) → (iii), (c) → (iv), (d) → (ii)

(B) (a) → (iv), (b) → (i), (c) → (ii), (d) → (iii)

(C) (a) → (ii), (b) → (iii), (c) → (iv), (d) → (i)

(D) (a) → (i), (b) → (iv), (c) → (iii), (d) → (ii)

Open solution

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Q5. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : A mixture of chloroform and aniline can be separated by simple distillation.

Statement II : When separating aniline from a mixture of aniline and water by steam distillation aniline boils below its boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Both Statement I and Statement II are false

(C) Statement I is true but Statement II is false

(D) Statement I is false but Statement II is true

Q6. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Dipole-dipole interactions are the only non-covalent interactions, resulting in hydrogen bond formation.

Reason R : Fluorine is the most electronegative element and hydrogen bonds in HF are symmetrical.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both A and R are true and R is the correct explanation of A

(B) A is true but R is false

(C) A is false but R is true

(D) Both A and R are true but R is NOT the correct explanation of A

Q7. MCQ (Single Correct Answer)

Open solution

Which one of the following lanthanoids does not form MO_2? [M is lanthanoid metal]

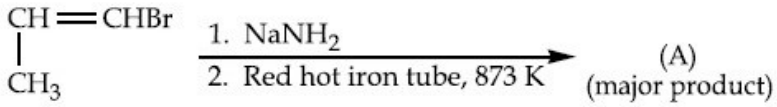
(A) Nd (B) Dy (C) Yb (D) Pr

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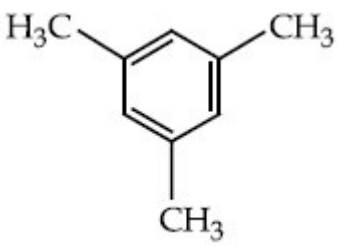
Q8. MCQ (Single Correct Answer)

Open solution

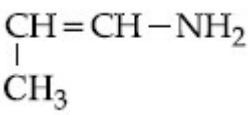
For the given reaction :



What is 'A'?

(A) 

(B) 

(C) 

(D) $CH_3CH_2CH_2NH_2$

Q9. MCQ (Single Correct Answer)

Open solution

Compound A used as a strong oxidizing agent is amphoteric in nature. It is the part of lead storage batteries. Compound A is :

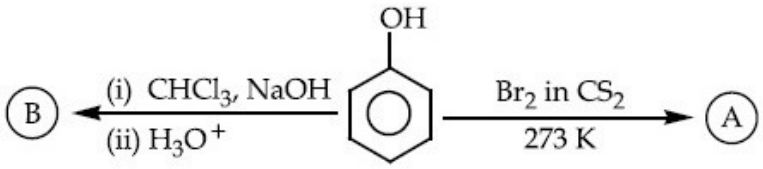
(A) PbO (B) PbO_2 (C) $PbSO_4$ (D) Pb_3O_4

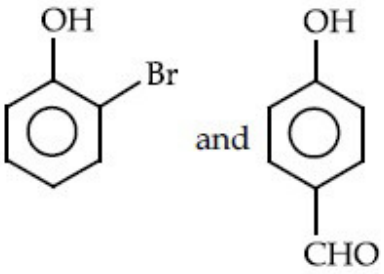
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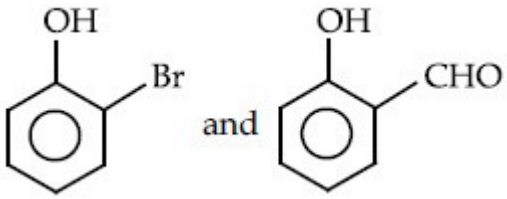
Q10. MCQ (Single Correct Answer)

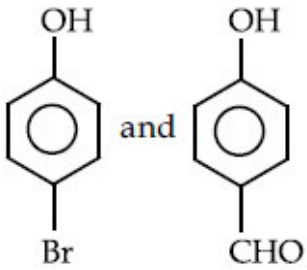
Open solution

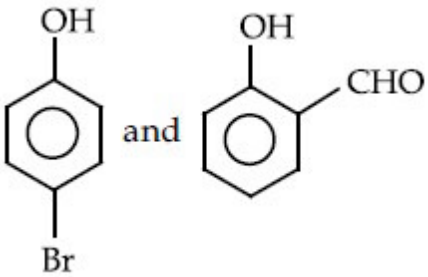
Identify the major products A and B respectively in the following reactions of phenol :



(A) 

(B) 

(C) 

(D) 

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Q11. MCQ (Single Correct Answer)

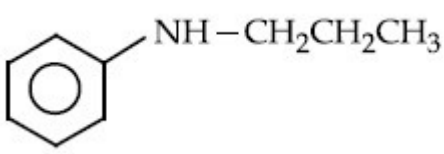
Open solution

An amine on reaction with benzenesulphonyl chloride produces a compound insoluble in alkaline solution. This amine can be prepared by ammonolysis of ethyl chloride. The correct structure of amine is :

(A) $CH_3CH_2NH_2$

(B) $CH_3CH_2CH_2NHCH_3$

(C) $CH_3CH_2CH_2\overset{H}{N}\text{-}CH_2CH_3$

(D) 

Q12. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : o-Nitrophenol is steam volatile due to intramolecular hydrogen bonding.

Statement II : o-Nitrophenol has high melting due to hydrogen bonding.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Both Statement I and Statement II are true

(B) Statement I is true but Statement II is false

(C) Statement I is false but Statement II is true

(D) Both Statement I and Statement II are false

Q13. MCQ (Single Correct Answer)

Open solution

$$\underset{(C_4H_8Cl_2)}{A} \xrightarrow[373\ K]{\text{Hydrolysis}} \underset{(C_4H_8O)}{B}$$

B reacts with Hydroxyl amine but does not give Tollen's test. Identify A and B.

(A) 1, 1-Dichlorobutane and Butanal

(B) 2, 2-Dichlorobutane and Butan-2-one

(C) 1, 1-Dichlorobutane and 2-Butanone

(D) 2, 2-Dichlorobutane and Butanal

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Q14. MCQ (Single Correct Answer)

Which of the following vitamins is helpful in delaying the blood clotting?

(A) Vitamin K (B) Vitamin C (C) Vitamin E (D) Vitamin B

Open solution

Q15. MCQ (Single Correct Answer)

The orbital having two radial as well as two angular nodes is :

(A) 3p (B) 5d (C) 4d (D) 4f

Open solution

Q16. Numerical

For a chemical reaction A + B ⇌ C + D

($\Delta_r H^\Theta$ = 80 kJ mol^{-1}) the entropy change $\Delta_r S^\Theta$ depends on the temperature T (in K) as $\Delta_r S^\Theta$ = 2T (J $K^{-1}mol^{-1}$).

Minimum temperature at which it will become spontaneous is _____ K. (Integer)

Open solution

Q17. Numerical

An exothermic reaction X $\rightarrow$ Y has an activation energy 30 kJ mol^{-1}. If energy change ΔE during the reaction is -20 kJ, then the activation energy for the reverse reaction in kJ is _____. (Integer answer)

Open solution

Q18. Numerical

Dichromate ion is treated with base, the oxidation number of Cr in the product formed is _____.

Open solution

Q19. Numerical

Number of bridging CO ligands in $[Mn_2(CO)_{10}]$ is _____.

Open solution

Q20. Numerical

Consider the following reaction

$$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{+2} + 4H_2O, E^o = 1.51V.$$

The quantity of electricity required in Faraday to reduce five moles of MnO_4^- is _____. (Integer answer)

Open solution

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Q21. Numerical

224 mL of $SO_{2(g)}$ at 298 K and 1 atm is passed through 100 mL of 0.1 M NaOH solution. The non-volatile solute produced is dissolved in 36g of water. The lowering of vapour pressure of solution (assuming the solution is dilute) ($P^o_{(H_2O)}$ = 24 mm of Hg) is $x \times 10^{-2}$ mm of Hg, the value of x is _____. (Integer answer)

Open solution

Q22. Numerical

A homogeneous ideal gaseous reaction $AB_{2(g)} \rightleftharpoons A_{(g)} + 2B_{(g)}$ is carried out in a 25 litre flask at 27°C. The initial amount of AB_2 was 1 mole and the equilibrium pressure was 1.9 atm. The value of K_p is $x \times 10^{-2}$. The value of x is _____. (Integer answer)

[R = 0.08206 $dm^3 atm\ K^{-1} mol^{-1}$]

Open solution

Mathematics 21 questions

Q23. MCQ (Single Correct Answer)

The maximum value of the term independent of 't' in the expansion of $\left(tx^{\frac{1}{5}} + \frac{(1-x)^{\frac{1}{10}}}{t}\right)^{10}$ where x∈(0, 1) is :

(A) $\frac{10!}{\sqrt{3}(5!)^2}$ (B) $\frac{2.10!}{3\sqrt{3}(5!)^2}$

(C) $\frac{10!}{3(5!)^2}$ (D) $\frac{2.10!}{3(5!)^2}$

Open solution

Q24. MCQ (Single Correct Answer)

The value of $\lim\limits_{h\to 0} 2\left\{\frac{\sqrt{3}\sin\left(\frac{\pi}{6}+h\right)-\cos\left(\frac{\pi}{6}+h\right)}{\sqrt{3}h\left(\sqrt{3}\cosh-\sinh\right)}\right\}$ is :

(A) $\frac{4}{3}$ (B) $\frac{2}{3}$ (C) $\frac{3}{4}$ (D) $\frac{2}{\sqrt{3}}$

Open solution

Q25. MCQ (Single Correct Answer)

Let A be a symmetric matrix of order 2 with integer entries. If the sum of the diagonal elements of A^2 is 1, then the possible number of such matrices is :

(A) 6 (B) 4 (C) 1 (D) 12

Open solution

Q26. MCQ (Single Correct Answer)

The value of $\int\limits_{-\pi/2}^{\pi/2} \frac{\cos^2 x}{1+3^x}\,dx$ is :

(A) 2π (B) $\frac{\pi}{2}$ (C) 4π (D) $\frac{\pi}{4}$

Open solution

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Q27. MCQ (Single Correct Answer)

The number of seven digit integers with sum of the digits equal to 10 and formed by using the digits 1, 2 and 3 only is :

(A) 35 (B) 42 (C) 82 (D) 77

Open solution

Q28. MCQ (Single Correct Answer)

Let R = {(P, Q) | P and Q are at the same distance from the origin} be a relation, then the equivalence class of (1, −1) is the set :

(A) $S = \{(x, y) | x^2 + y^2 = \sqrt{2}\}$ (B) $S = \{(x, y) | x^2 + y^2 = 2\}$

(C) $S = \{(x, y) | x^2 + y^2 = 1\}$ (D) $S = \{(x, y) | x^2 + y^2 = 4\}$

Open solution

Q29. MCQ (Single Correct Answer)

The value of $\sum_{n=1}^{100} \int_{n-1}^{n} e^{x-[x]} dx$, where [x] is the greatest integer $\leq$ x, is :

(A) 100e (B) 100(e − 1) (C) 100(1 + e) (D) 100(1 − e)

Open solution

Q30. MCQ (Single Correct Answer)

The intersection of three lines x − y = 0, x + 2y = 3 and 2x + y = 6 is a :

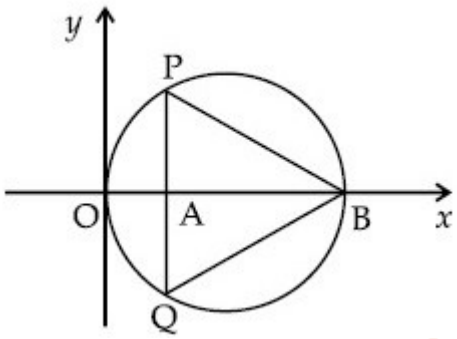
(A) Right angled triangle (B) Equilateral triangle

(C) None of the above (D) Isosceles triangle

Open solution

Q31. MCQ (Single Correct Answer)

In the circle given below, let OA = 1 unit, OB = 13 unit and PQ $\perp$ OB. Then, the area of the triangle PQB (in square units) is :



(A) $24\sqrt{2}$ (B) $24\sqrt{3}$ (C) $26\sqrt{2}$ (D) $26\sqrt{3}$

Open solution

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Q32. MCQ (Single Correct Answer)

The value of $\begin{vmatrix}(a+1)(a+2) & a+2 & 1\\(a+2)(a+3) & a+3 & 1\\(a+3)(a+4) & a+4 & 1\end{vmatrix}$ is :

(A) -2 (B) 0

(C) (a + 2)(a + 3)(a + 4) (D) (a + 1)(a + 2)(a + 3)

Open solution

Q33. MCQ (Single Correct Answer)

If $\frac{\sin^{-1}x}{a} = \frac{\cos^{-1}x}{b} = \frac{\tan^{-1}y}{c}; 0 < x < 1,$
then the value of $\cos\left(\frac{\pi c}{a+b}\right)$ is :

(A) $\frac{1-y^2}{2y}$ (B) $\frac{1-y^2}{y\sqrt{y}}$ (C) $1-y^2$ (D) $\frac{1-y^2}{1+y^2}$

Open solution

Q34. MCQ (Single Correct Answer)

The maximum slope of the curve $y = \frac{1}{2}x^4 - 5x^3 + 18x^2 - 19x$ occurs at the point :

(A) $\left(3, \frac{21}{2}\right)$ (B) (0, 0)

(C) (2, 9) (D) (2, 2)

Open solution

Q35. MCQ (Single Correct Answer)

The rate of growth of bacteria in a culture is proportional to the number of bacteria present and the bacteria count is 1000 at initial time t = 0. The number of bacteria is increased by 20% in 2 hours. If the population of bacteria is 2000 after $\frac{k}{\log_e\left(\frac{6}{5}\right)}$ hours, then $\left(\frac{k}{\log_e 2}\right)^2$ is equal to :

(A) 16 (B) 8 (C) 2 (D) 4

Open solution

Q36. MCQ (Single Correct Answer)

In an increasing geometric series, the sum of the second and the sixth term is $\frac{25}{2}$ and the product of the third and fifth term is 25. Then, the sum of 4th, 6th and 8th terms is equal to :

(A) 30 (B) 32 (C) 26 (D) 35

Open solution

Q37. Numerical

The value of the integral $\int_0^{\pi} |\sin 2x| dx$ is _____.

Open solution

Q38. Numerical

The number of solutions of the equation $\log_4(x-1) = \log_2(x-3)$ is _____.

Open solution

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Q39. Numerical

The sum of 162th power of the roots of the equation $x^3 - 2x^2 + 2x - 1 = 0$ is _____.

Open solution

Q40. Numerical

The area bounded by the curve $y = ||x - 1| - 2|$ and the line $y = 2$ is _____.

Open solution

Q41. Numerical

The difference between degree and order of a differential equation that represents the family of curves given by $y^2 = a\left(x + \frac{\sqrt{a}}{2}\right)$, a > 0 is _____.

Open solution

Q42. Numerical

If y = y(x) is the solution of the equation

$e^{\sin y}\cos y\frac{dy}{dx} + e^{\sin y}\cos x = \cos x$, y(0) = 0; then

$1 + y\left(\frac{\pi}{6}\right) + \frac{\sqrt{3}}{2}y\left(\frac{\pi}{3}\right) + \frac{1}{\sqrt{2}}y\left(\frac{\pi}{4}\right)$ is equal to _____.

Open solution

Q43. Numerical

The number of integral values of 'k' for which the equation $3\sin x + 4\cos x = k + 1$ has a solution, $k \in R$ is _____.

Open solution

Physics 28 questions

Q44. MCQ (Single Correct Answer)

Consider the combination of 2 capacitors C_1 and C_2 with $C_2 > C_1$, when connected in parallel, the equivalent capacitance is $\frac{15}{4}$ times the equivalent capacitance of the same connected in series. Calculate the ratio of capacitors, $\frac{C_2}{C_1}$.

Open solution

(A) $\frac{15}{11}$ (B) No Solutions (C) $\frac{29}{15}$ (D) $\frac{15}{4}$

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Q45. MCQ (Single Correct Answer)

An alternating current is given by the equation i = i_1 sin ωt + i_2 cos ωt. The rms current will be :

(A) $\frac{1}{\sqrt{2}}\left(i_1^2+i_2^2\right)^{\frac{1}{2}}$

(B) $\frac{1}{\sqrt{2}}(i_1+i_2)$

(C) $\frac{1}{\sqrt{2}}(i_1+i_2)^2$

(D) $\frac{1}{2}\left(i_1^2+i_2^2\right)^{\frac{1}{2}}$

Open solution

Q46. MCQ (Single Correct Answer)

Four identical solid spheres each of mass 'm' and radius 'a' are placed with their centres on the four corners of a square of side 'b'. The moment of inertia of the system about one side of square where the axis of rotation is parallel to the plane of the square is :

(A) $\frac{4}{5}ma^2$

(B) $\frac{8}{5}ma^2+mb^2$

(C) $\frac{4}{5}ma^2+2mb^2$

(D) $\frac{8}{5}ma^2+2mb^2$

Open solution

Q47. MCQ (Single Correct Answer)

A large number of water drops, each of radius r, combine to have a drop of radius R. If the surface tension is T and mechanical equivalent of heat is J, the rise in heat energy per unit volume will be :

(A) $\frac{2T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

(B) $\frac{3T}{J}\left(\frac{1}{r}-\frac{1}{R}\right)$

(C) $\frac{3T}{rJ}$

(D) $\frac{2T}{rJ}$

Open solution

Q48. MCQ (Single Correct Answer)

LED is constructed from Ga-As-P semiconducting material. The energy gap of this LED is 1.9 eV. Calculate the wavelength of light emitted and its colour.

[h = 6.63 $\times$ 10^{-34} Js and c = 3 $\times$ 10^8 ms^{-1}]

(A) 654 nm and orange colour

(B) 654 nm and red colour

(C) 1046 nm and red colour

(D) 1046 nm and blue colour

Open solution

Q49. MCQ (Single Correct Answer)

A short straight object of height 100 cm lies before the central axis of a spherical mirror whose focal length has absolute value | f | = 40 cm. The image of object produced by the mirror is of height 25 cm and has the same orientation of the object. One may conclude from the information :

(A) Image is real, same side of convex mirror.

(B) Image is virtual, opposite side of convex mirror.

(C) Image is virtual, opposite side of concave mirror.

(D) Image is real, same side of concave mirror.

Open solution

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Q50. MCQ (Single Correct Answer)

In a Young's double slit experiment two slits are separated by 2 mm and the screen is placed one meter away. When a light of wavelength 500 nm is used, the fringe separation will be :

(A) 0.50 mm (B) 0.25 mm (C) 1 mm (D) 0.75 mm

Open solution

Q51. MCQ (Single Correct Answer)

Assume that a tunnel is dug along a chord of the earth, at a perpendicular distance (R/2) from the earth's centre, where 'R' is the radius of the Earth. The wall of the tunnel is frictionless. If a particle is released in this tunnel, it will execute a simple harmonic motion with a time period :

(A) $2\pi\sqrt{\frac{R}{g}}$ (B) $\frac{g}{2\pi R}$

(C) $\frac{2\pi R}{g}$ (D) $\frac{1}{2\pi}\sqrt{\frac{g}{R}}$

Open solution

Q52. MCQ (Single Correct Answer)

In a typical combustion engine the workdone by a gas molecule is given by $W = \alpha^2\beta e^{\frac{-\beta x^2}{kT}}$, where x is the displacement, k is the Boltzmann constant and T is the temperature. If α and β are constants, dimensions of α will be :

(A) $[M^0LT^0]$ (B) $[MLT^{-1}]$ (C) $[MLT^{-2}]$ (D) $[M^2LT^{-2}]$

Open solution

Q53. MCQ (Single Correct Answer)

If two similar springs each of spring constant K_1 are joined in series, the new spring constant and time period would be changed by a factor :

(A) $\frac{1}{2}, 2\sqrt{2}$ (B) $\frac{1}{4}, 2\sqrt{2}$ (C) $\frac{1}{2}, \sqrt{2}$ (D) $\frac{1}{4}, \sqrt{2}$

Open solution

Q54. MCQ (Single Correct Answer)

A particle is moving with uniform speed along the circumference of a circle of radius R under the action of a central fictitious force F which is inversely proportional to R^3. Its time period of revolution will be given by :

(A) $T \propto R^{\frac{4}{3}}$ (B) $T \propto R^{\frac{5}{2}}$ (C) $T \propto R^{\frac{3}{2}}$ (D) $T \propto R^2$

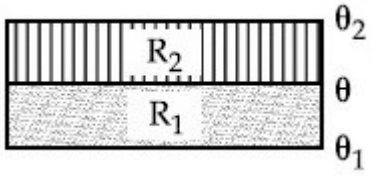
Open solution

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Q55. MCQ (Single Correct Answer)

The temperature θ at the junction of two insulating sheets, having thermal resistances R_1 and R_2 as well as top and bottom temperatures θ_1 and θ_2 (as shown in figure) is given by :

Open solution



(A) $\frac{\theta_1 R_2+\theta_2 R_1}{R_1+R_2}$

(B) $\frac{\theta_1 R_1+\theta_2 R_2}{R_1+R_2}$

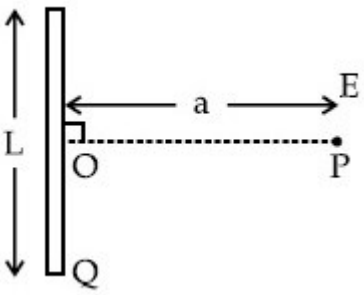
(C) $\frac{\theta_1 R_2-\theta_2 R_1}{R_2-R_1}$

(D) $\frac{\theta_2 R_2-\theta_1 R_1}{R_2-R_1}$

Q56. MCQ (Single Correct Answer)

Find the electric field at point P (as shown in figure) on the perpendicular bisector of a uniformly charged thin wire of length L carrying a charge Q. The distance of the point P from the centre of the wire is a = $\frac{\sqrt{3}}{2}L$.

Open solution



(A) $\frac{Q}{4\pi\varepsilon_0 L^2}$

(B) $\frac{Q}{3\pi\varepsilon_0 L^2}$

(C) $\frac{Q}{2\sqrt{3}\pi\varepsilon_0 L^2}$

(D) $\frac{\sqrt{3}Q}{4\pi\varepsilon_0 L^2}$

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Q57. MCQ (Single Correct Answer)

Open solution

A planet revolving in elliptical orbit has :

A. a constant velocity of revolution.

B. has the least velocity when it is nearest to the sun.

C. its areal velocity is directly proportional to its velocity.

D. areal velocity is inversely proportional to its velocity.

E. to follow a trajectory such that the areal velocity is constant.

Choose the correct answer from the options given below :

(A) D only **(B)** E only **(C)** C only **(D)** A only

Q58. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labeled as Assertion A and the other is labelled as Reason R.

Assertion A : An electron microscope can achieve better resolving power than an optical microscope.

Reason R : The de Broglie's wavelength of the electrons emitted from an electron gun is much less than wavelength of visible light.

In the light of the above statements, choose the correct answer from the options given below :

(A) A is false but R is true.

(B) Both A and R are true but R is NOT the correct explanation of A.

(C) Both A and R are true and R is the correct explanation of A.

(D) A is true but R is false.

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Q59. MCQ (Single Correct Answer)

Open solution

Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R.

Assertion A : Body 'P' having mass M moving with speed 'u' has head-on collision elastically with another body 'Q' having mass 'm' initially at rest. If m << M, body 'Q' will have a maximum speed equal to '2u' after collision.

Reason R : During elastic collision, the momentum and kinetic energy are both conserved.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) A is correct but R is not correct.

(B) A is not correct but R is correct.

(C) Both A and R are correct and R is the correct explanation of A.

(D) Both A and R are correct but R is NOT the correct explanation of A.

Q60. MCQ (Single Correct Answer)

Open solution

The normal density of a material is ρ and its bulk modulus of elasticity is K. The magnitude of increase in density of material, when a pressure P is applied uniformly on all sides, will be :

(A) $\frac{\rho K}{P}$ (B) $\frac{PK}{\rho}$ (C) $\frac{\rho P}{K}$ (D) $\frac{K}{\rho P}$

Q61. MCQ (Single Correct Answer)

Open solution

If λ_1 and λ_2 are the wavelengths of the third member of Lyman and first member of the Paschen series respectively, then the value of $\lambda_1 : \lambda_2$ is :

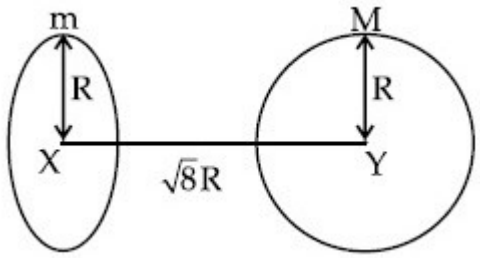
(A) 7 : 135 (B) 7 : 108 (C) 1 : 9 (D) 1 : 3

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Q62. MCQ (Single Correct Answer)

Open solution

Find the gravitational force of attraction between the ring and sphere as shown in the diagram, where the plane of the ring is perpendicular to the line joining the centres. If $\sqrt{8}$R is the distance between the centres of a ring (of mass 'm') and a sphere (mass 'M') where both have equal radius 'R'.



(A) $\frac{2\sqrt{2}}{3}.\frac{GMm}{R^2}$

(B) $\frac{\sqrt{8}}{9}.\frac{GmM}{R}$

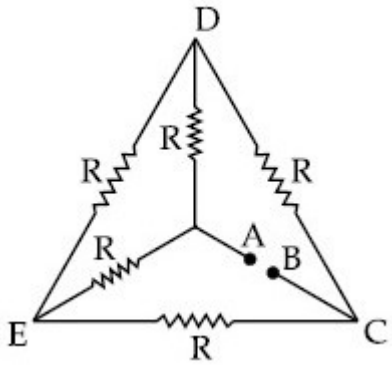
(C) $\frac{\sqrt{8}}{27}.\frac{GmM}{R^2}$

(D) $\frac{1}{3\sqrt{8}}.\frac{GMm}{R^2}$

Q63. MCQ (Single Correct Answer)

Open solution

Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is :



(A) $\frac{3R}{2}$

(B) $\frac{R}{2}$

(C) 2R

(D) R

Q64. Numerical

Open solution

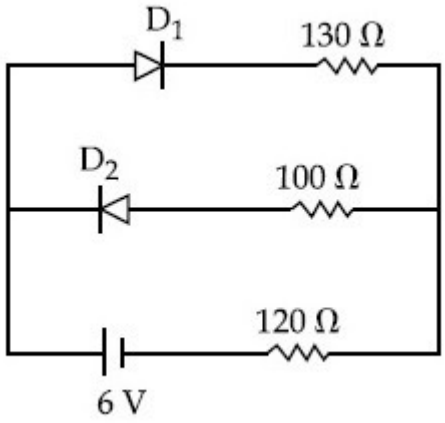
A person standing on a spring balance inside a stationary lift measures 60 kg. The weight of that person if the lift descends with uniform downward acceleration of 1.8 m/s^2 will be _____ N. [g = 10 m/s^2]

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Q65. Numerical

The circuit contains two diodes each with a forward resistance of 50Ω and with infinite reverse resistance. If the battery voltage is 6V, the current through the 120Ω resistance is _____ mA.

Open solution



Q66. Numerical

The mass per unit length of a uniform wire is 0.135 g/cm. A transverse wave of the form y = − 0.21 sin (x + 30t) is produced in it, where x is in meter and t is in second. Then, the expected value of tension in the wire is x $\times$ 10^{-2} N. Value of x is _____. (Round off to the nearest integer)

Open solution

Q67. Numerical

In an electrical circuit, a battery is connected to pass 20C of charge through it in a certain given time. The potential difference between two plates of the battery is maintained at 15V. The workdone by the battery is _____J.

Open solution

Q68. Numerical

A boy pushes a box of mass 2 kg with a force $\vec{F} = \left(20\hat{i} + 10\hat{j}\right) N$ on a frictionless surface. If the box was initially at rest, then _____ m is displacement along the x-axis after 10s.

Open solution

Q69. Numerical

A radiation is emitted by 1000W bulb and it generates an electric field and magnetic field at P, placed at a distance of 2m. The efficiency of the bulb is 1.25%. The value of peak electric field at P is x $\times$ 10^{-1} V/m. Value of x is _____. (Rounded off to the nearest integer) [Take $\varepsilon_0 = 8.85 \times 10^{-12}$ C^2N^{-1} m^{-2}, c = 3×10^8 ms^{-1}]

Open solution

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Q70. Numerical

A container is divided into two chambers by a partition. The volume of first chamber is 4.5 litre and second chamber is 5.5 litre. The first chamber contain 3.0 moles of gas at pressure 2.0 atm and second chamber contain 4.0 moles of gas at pressure 3.0 atm. After the partition is removed and the mixture attains equilibrium, then, the common equilibrium pressure existing in the mixture is x $\times$ 10^{-1} atm. Value of x is _____.

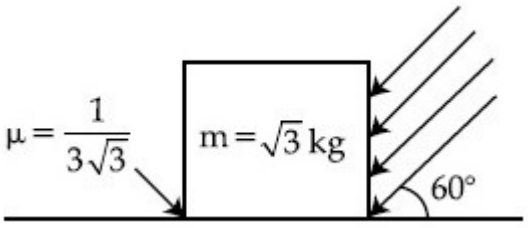
Open solution

Q71. Numerical

As shown in the figure, a block of mass $\sqrt{3}$ kg is kept on a horizontal rough surface of coefficient of friction $\frac{1}{3\sqrt{3}}$. The critical force to be applied on the vertical surface as shown at an angle 60° with horizontal such that it does not move, will be 3x. The value of x will be _____.

Open solution

[g = 10 m/s^2; sin60° = $\frac{\sqrt{3}}{2}$; cos60° = $\frac{1}{2}$]



Out of syllabus 19 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q72. MCQ (Single Correct Answer) · **Chemistry**

Open solution

Statements about heavy water are given below :

A. Heavy water is used in exchange reactions for the study of reaction mechanisms.

B. Heavy water is prepared by exhaustive electrolysis of water.

(C) Heavy water has higher boiling point than ordinary water.

(D) Viscosity of H_2O is greater than D_2O.

Choose the most appropriate answer from the options given below :

(A) A, B, and C only **(B)** A and C only **(C)** A and D only **(D)** A and B only

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Q73. MCQ (Single Correct Answer) · **Chemistry**

The presence of ozone in troposphere :

(A) protects us from greenhouse effect

(B) protects us from the UV radiation

(C) protects us from the X-ray radiation

(D) generates photochemical smog

Open solution

Q74. MCQ (Single Correct Answer) · **Chemistry**

Match List - I with List - II.

	List - I (Ore)		List - II (Element Present)
(a)	Kernite	(i)	Tin
(b)	Cassiterite	(ii)	Boron
(c)	Calamine	(iii)	Fluorine
(d)	Cryolite	(iv)	Zinc

Choose the most appropriate answer from the options given below :

(A) (a) → (i), (b) → (iii), (c) → (iv), (d) → (ii)

(B) (a) → (iii), (b) → (i), (c) → (ii), (d) → (iv)

(C) (a) → (ii), (b) → (i), (c) → (iv), (d) → (iii)

(D) (a) → (ii), (b) → (iv), (c) → (i), (d) → (iii)

Open solution

Q75. MCQ (Single Correct Answer) · **Chemistry**

On treating a compound with warm dil. H_2SO_4, gas X is evolved which turns $K_2Cr_2O_7$ paper acidified with dil. H_2SO_4 to a green compound Y. X and Y respectively are :

(A) X = SO_2, Y = Cr_2O_3

(B) X = SO_3, Y = $Cr_2(SO_4)_3$

(C) X = SO_3, Y = Cr_2O_3

(D) X = SO_2, Y = $Cr_2(SO_4)_3$

Open solution

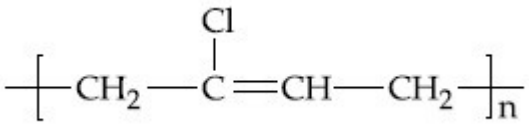
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Q76. MCQ (Single Correct Answer) · **Chemistry**

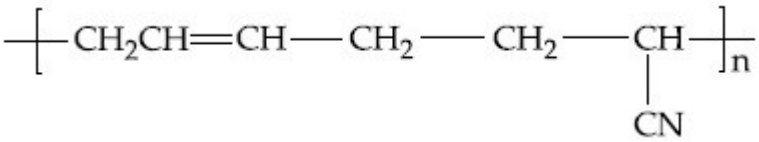
Open solution

The structure of Neoprene is :

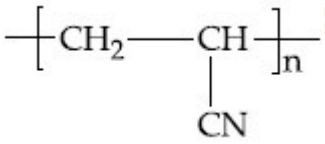
(A)



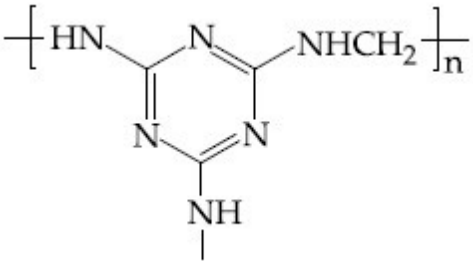
(B)



(C)



(D)



Q77. Numerical · **Chemistry**

Open solution

3.12 g of oxygen is adsorbed on 1.2 g of platinum metal. The volume of oxygen adsorbed per gram of the adsorbent at 1 atm and 300 K in L is _____.

[R = 0.0821 L atm $K^{-1}mol^{-1}$]

Q78. Numerical · **Chemistry**

Open solution

The number of significant figures in 50000.020×10^{-3} is _____.

Q79. Numerical · **Chemistry**

Open solution

A certain gas obeys $P(V_m - b) = RT$. The value of $\left(\frac{\partial Z}{\partial P}\right)_T$ is $\frac{xb}{RT}$. The value of x is _____. (Integer answer) (Z : compressibility factor)

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Q80. MCQ (Single Correct Answer) · **Mathematics**

The sum of the infinite series
$1+\frac{2}{3}+\frac{7}{3^2}+\frac{12}{3^3}+\frac{17}{3^4}+\frac{22}{3^5}+\ldots\ldots$ is equal to :

(A) $\frac{9}{4}$ (B) $\frac{13}{4}$ (C) $\frac{15}{4}$ (D) $\frac{11}{4}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If $\vec{a}$ and $\vec{b}$ are perpendicular, then
$\vec{a}\times\left(\vec{a}\times\left(\vec{a}\times\left(\vec{a}\times\vec{b}\right)\right)\right)$ is equal to :

(A) $\frac{1}{2}|\vec{a}|^4\vec{b}$

(B) $\vec{0}$

(C) $\vec{a}\times\vec{b}$

(D) $|\vec{a}|^4\vec{b}$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

A fair coin is tossed a fixed number of times. If the probability of getting 7 heads is equal to probability of getting 9 heads, then the probability of getting 2 heads is :

(A) $\frac{15}{2^8}$ (B) $\frac{15}{2^{12}}$ (C) $\frac{15}{2^{13}}$ (D) $\frac{15}{2^{14}}$

Open solution

Q83. MCQ (Single Correct Answer) · **Mathematics**

Consider the three planes

$P_1 : 3x + 15y + 21z = 9,$

$P_2 : x - 3y - z = 5,$ and

$P_3 : 2x + 10y + 14z = 5$

Then, which one of the following is true?

(A) P_1 and P_2 are parallel. (B) P_1, P_2 and P_3 all are parallel.

(C) P_1 and P_3 are parallel. (D) P_2 and P_3 are parallel.

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

If (1, 5, 35), (7, 5, 5), (1, λ, 7) and (2λ, 1, 2) are coplanar, then the sum of all possible values of λ is :

(A) $-\frac{44}{5}$ (B) $-\frac{39}{5}$ (C) $\frac{44}{5}$ (D) $\frac{39}{5}$

Open solution

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Q85. Numerical · **Mathematics**

Let $(\lambda, 2, 1)$ be a point on the plane which passes through the point $(4, -2, 2)$. If the plane is perpendicular to the line joining the points $(-2, -21, 29)$ and $(-1, -16, 23)$, then $\left(\frac{\lambda}{11}\right)^2 - \frac{4\lambda}{11} - 4$ is equal to _____.

Open solution

Q86. Numerical · **Mathematics**

If $\sqrt{3}(\cos^2 x) = (\sqrt{3} - 1)\cos x + 1$, the number of solutions of the given equation when $x \in \left[0, \frac{\pi}{2}\right]$ is _____.

Open solution

Q87. Numerical · **Mathematics**

Let m, n∈N and gcd (2, n) = 1. If $30\binom{30}{0} + 29\binom{30}{1} + \ldots\ldots + 2\binom{30}{28} + 1\binom{30}{29} = n.2^m$, then n + m is equal to _____.

(Here $\binom{n}{k} = {}^nC_k$)

Open solution

Q88. MCQ (Single Correct Answer) · **Mathematics**

Let f be any function defined on R and let it satisfy the condition :
$|f(x) - f(y)| \leq |(x-y)^2|, \forall (x, y) \in R$

If f(0) = 1, then :

(A) f(x) can take any value in R

(B) $f(x) < 0, \forall x \in R$

(C) $f(x) > 0, \forall x \in R$

(D) $f(x) = 0, \forall x \in R$

Open solution

Q89. Numerical · **Physics**

In a series LCR resonant circuit, the quality factor is measured as 100. If the inductance is increased by two fold and resistance is decreased by two fold, then the quality factor after this change will be _____.

Open solution

Q90. Numerical · **Physics**

The maximum and minimum amplitude of an amplitude modulated wave is 16V and 8V respectively. The modulation index for this amplitude modulated wave is x $\times 10^{-2}$. The value of x is _____.

Open solution

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Answer Key

Chemistry

1. **A**	2. **A**	3. **D**	4. **C**	5. **A**
6. **C**	7. **C**	8. **A**	9. **B**	10. **D**
11. **C**	12. **B**	13. **B**	14. **A**	15. **B**
16. **200**	17. **50**	18. **6**	19. **0**	20. **25**
21. **18 to 24**	22. **72 to 75**			

Mathematics

23. **B**	24. **A**	25. **B**	26. **D**	27. **D**
28. **B**	29. **B**	30. **D**	31. **B**	32. **A**
33. **D**	34. **D**	35. **D**	36. **D**	37. **2**
38. **1**	39. **3**	40. **8**	41. **2**	42. **1**
43. **11**				

Physics

44. **B**	45. **A**	46. **D**	47. **B**	48. **B**
49. **B**	50. **B**	51. **A**	52. **A**	53. **C**
54. **D**	55. **A**	56. **C**	57. **B**	58. **C**
59. **C**	60. **C**	61. **A**	62. **C**	63. **D**
64. **492**	65. **20**	66. **1215**	67. **300**	68. **500**
69. **137**	70. **25**	71. **3.33**		

Out of syllabus

72. **A**	73. **D**	74. **C**	75. **D**	76. **A**
77. **2**	78. **8**	79. **1**	80. **B**	81. **D**
82. **C**	83. **C**	84. **C**	85. **8**	86. **1**
87. **45**	88. **C**	89. **283**	90. **33**	

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

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